

Recall AI Use-Case Specification

Version 1.0

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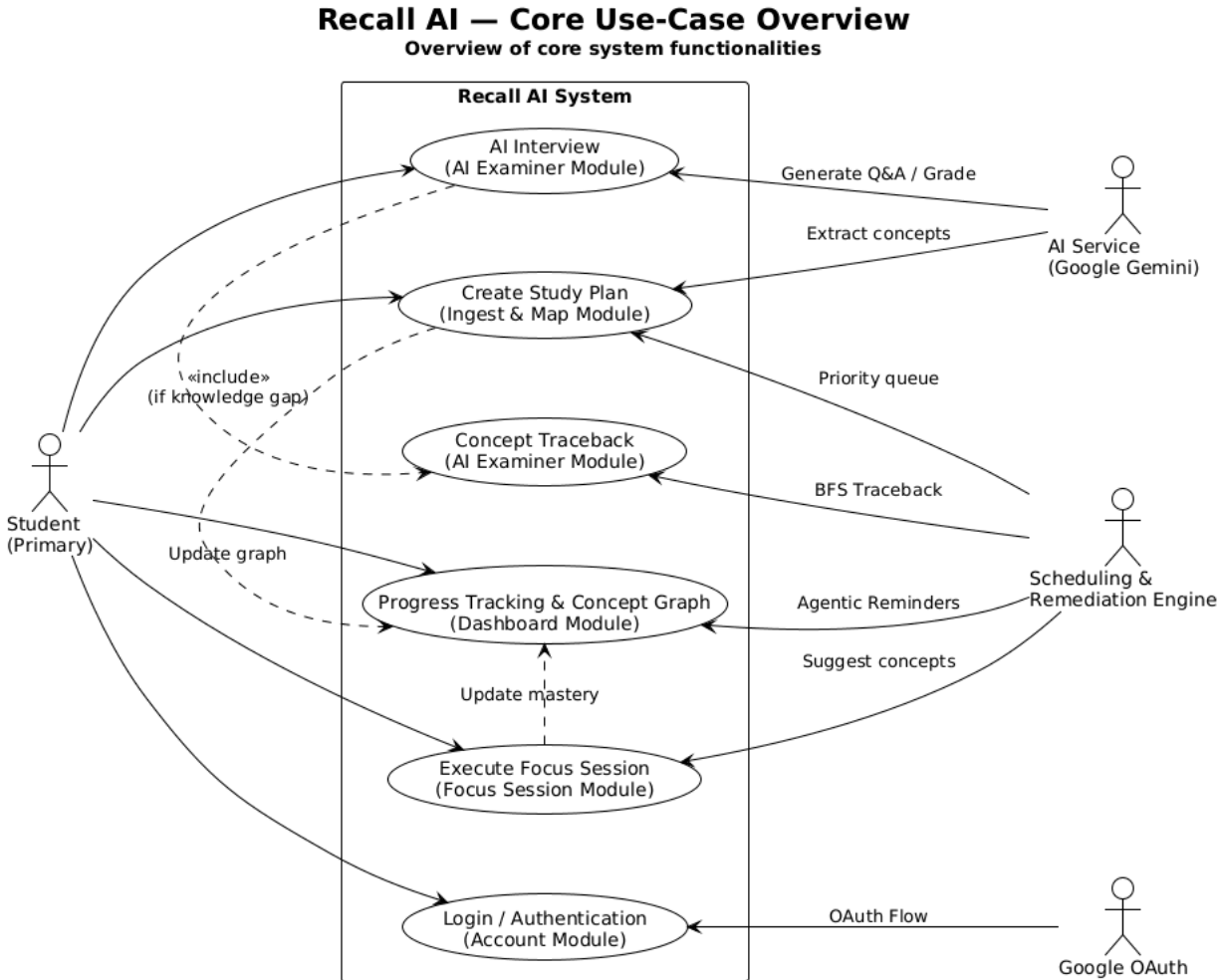
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1. Use-case Model



2. Use-case Specifications

2.1 Use-case: Create Study Plan

Use case Name	Create Study Plan
Brief description	This use case allows a Student to create a new study plan by uploading study materials (text, images, PDF). The system will automatically call the AI Service to extract concepts and build a prerequisite relationship graph between them. After the user reviews and customizes the graph, the Scheduling & Remediation Engine will initialize an initial study schedule based on the concept graph and the set deadline
Actors	- Student - AI Service - Scheduling & Remediation Engine (SRE)

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Basic Flow	1. The Student selects the “Create study plan” function on the system the system interface (Dashboard) 2. The System displays a form requesting study plan information, including: subject/plan name, deadline, and a document upload area. 3. The Student fill in the necessary information, uploads the study material (supporting Text, Image, PDF formats), and clicks the “Create” button. 4. The System receives the data and sends the document to the AI Service with a request to extract the list of concepts (with difficulty) and prerequisite relationships. 5. The AI Service analyzes the document and return the result to the system in a fixed-structure JSON format. 6. The System receives the JSON data and proceeds to validate the graph’s validity (Validate DAG - Directed Acyclic Graph) to ensure there are no cycles between concepts. 7. The System displays the newly extracted visual graph (concept map) on the screen and asks the Student to review it (This step includes the Use case <i>Interact with Plan Graph</i>). 8. The Student reviews, can add/remove node/edges if necessary. Every time a new edge is added, the System performs a real-time cycle check (DAG validation). If a cycle is detected, the edit is rejected with a warning. After all edits, the Student confirm the completion of the graph. 9. The System saves the plan information, concepts (concepts), and graph relationships (concept_edges) into the database. 10. The System calls the Scheduling & Remediation Engine (SRE) to generate an initial study schedule based on the finalized concept graph and the Student’s deadline. 11. The System displays a success message for plan creation and redicrects the Study to the Dashboard to view progress.
Alternative Flows	Alternative flow 1: Invalid input material 1. From #3 of the basic flow, if the uploaded document is not in a supported format, is blurry/unreadable (for images), or exceed ithe allowed file size, the system display an error message directly on the form and asks the Student to reselect the document. 2. The Student reselects the document. 3. Continue step #3 in the basic flow Alternative flow 2: AI Service return invalid JSON format 1. From #5 of the basic flow, if the AI Service returns results as free text or malformed JSON, the system automatically retries sending the request or splits the document by headings/paragraphs for processing. 2. If the maximum number of retries is reached and it still fails, the system reports an automatic extraction error and allows the Student to manually input concepts. 3. Continue step #8 in the basic flow (Sudent manually inputs concepts)

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	Alternative flow 3: Generated graph constraints cycles 1. From #6 of the basic flow, if the system detects that the extracted graph contains cycles (is not a DAG), it automatically removes the edge causing the cycle and log a warning. 2. Continue step #7 in the basic flow for the Student to manually check. Alternative flow 4: AI Service encounters an error or is overloaded 1. From #5 of the basic flow, if the AI Service does not respond (timeout) or the system runs out of API quota, the system displays the error: "AI analysis service is interrupted, please try again later" 2. The use case terminates Alternatice flow 5: Student cancels the plan creation 1. At any point before step #8 of the basic flow, if the Student clicks "Cancel" or leaves the page, the system deletes the temporary unsaved data. 2. The use case terminates.
Pre-conditions	- The Student has successfully logged into the Recall AI system. - The AI Service is in a normal operating state, and the system has sufficient API quota/rate limit
Post-conditions	- A new Study Plan record is successfully saved in the database. - The concepts and corresponding prerequisite relationships of the plan are fully stored as a Directed Acyclic Graph (DAG) - An initial study schedule for the plan has created and is ready for study sessions (Focus Session/Interview).

2.2 Use-case: Execute Focus Session

Use case Name	Execute Focus Session
Brief description	This use case describes the process of a Student executing a time-managed study session (e.g., Pomodoro). During the session, the Student can view the original document (PDF) side-by-side and take quick notes related to the concepts being studied. Upon completion, the system saves the session record (time studied) and interacts with the Scheduling & Remediation Engine (SRE) to update the progress (mastery score), preparing for the subsequent knowledge verification step.
Actor	- Student - Scheduling & Remediation Engine (SRE)
Basic Flow	1. Student requests to start a new study session from the application interface (Dashboard or subject details screen). 2. System displays the setup screen, asking the Student to select a concept

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	to review and configure the study time (default to Pomodoro). Student selects the desired concept and confirms to start. System displays the main session interface, including the countdown timer, side-by-side original PDF document view, and the note-taking tool Student clicks the “Start” button. System starts the countdown timer. Student read the document, studies, and enters quick notes linked to the concept. System automatically auto-saves the Student’s notes. - When the timer finishes counting down (or Student manually click to end), the Student selects “End session”. System saves the entire session record (total time, concepts, notes) and send the information to the SRE. SRE (Actor) receives the information, calculates, and updates the corresponding mastery parameters of the concept. System displays the session completion screen with a summary of the results, providing subsequent options (e.g., take a break or move to AI Examiner).
Alternative Flows	Alternative flow 1: View and select Concept Recommendations (Extension) - From step #2 of the basic flow, if the Student selects the “View Recommendations” feature. - System sends a request to the SRE. SRE calculates and provides a list of priority concepts (based on weak mastery or upcoming deadlines). System displays the list of recommendations to the Student and Student selects a concept from the recommended list. - Continue step #3. Alternative flow 2: Enable Strict Mode (Extension) - From step #4 or #6 of the basic flow, if the Student selects to enable “Strict Mode”. - System activates the mechanism to block/restrict distracting websites or tabs. - Continue step #4 (or #6) Alternative flow 3: Pause and Resume - From step #6 of the basic flow, if the Student clicks the “Pause” button on the timer. - System pauses the countdown timer and saves the current state. The Student clicks “Resume”. System resumes the countdown timer. - Continue step #6 Alternative flow 4: Cancel the session midway - From step #5 to step #8 of the basic flow, if the Student selects “Candle

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	session” 2. System displays a dialog asking to confirm the cancellation. The Student confirms the cancellation. System closes the session, does not record the study time (or saves it “Cancelled” status), and redirects the user to previous interface. 3. The use case terminates
Pre-conditions	- The Student has successfully logged into the Recall AI system. - The Student has created a Study Plan, and the system already has a list of concepts along with their corresponding prerequisite graph.
Post-conditions	- The study duration and session status are successfully saved to the system history. - The student’s quick notes are saved and correctly linked to the selected concepts. - The SRE record the session completion data to update mastery parameters (study statistics)

2.3 Use-case: Execute Interview Session

Use case Name	Execute Interview Session (Multi-turn Q&A)
Brief description	This use case allows a Student to take a knowledge test in the form of a multi-turn Q&A interview. The system use AI to act as an examiner, generating questions and grading answers directly based on the original materials uploaded by the Student. The workflow is controlled by a state machine that limits the number of Q&A turns, helping to classify the actual level of comprehension and automatically triggering a prerequisite traceback if it detects the Student has gaps in foundational knowledge.
Actor	- Student - AI Service
Basic Flow	1. Student starts the interview session from the queue session of concepts to review. 2. System loads the highest priority concept, sets the limit for Q&A turns (e.g., 3 turns), and sends a request (generate_question) to the AI Service 3. AI Service generates an interview question based on the rubric and original document, and simultaneously generates audio data (Text-to-Speech). 4. System plays the AI voice audio for the Student to hear (voice interaction). The text of the question is hidden by default (only automatically displayed

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	if the question has a complex structure, or when the student actively click the “Show question” button). 5. Student answers the question using Voice (Speech) through the device’s microphone. 6. System records the audio, automatically converts it to text (Speech-to-Text) so the Student can quickly review the content. 7. System sends the sends the text answer to the AI Service for comparison and grading (grade_answer). 8. AI Service analyzes and returns the score, feedback, and the level of understanding classification 9. System processes the AI’s feedback: - If the answer shows deep understanding, the system asks the AI to create more profound application question (loops back to step 3) - If the answer is superficial, the system asks the AI to pose a follow-up question forcing the Student to explain more clearly (loops back to step 3) - If the answer is wrong/missing knowledge, the AI briefly explains the mistake. The system immediately to find the root concept 10. System saves the evaluation result of the current concept after reaching the maximum number of turns or finishing the processing loop in step 9. 11. System repeats the process from Step 2 for the next concept in the queue until the list is completed. 12. System announces the end of the interview session via voice and automatically redirects the Student to the End-of-Session Summary screen.
Alternative Flows	Alternative flow 1: AI Service Timeout or Error (AI Fail/Timeout) 1. From #3 or #7 of the basic flow, if the AI Service responds to slowly, runs out of quota, or reports an error. 2. The system log the error and automatically Fallback. The system stop calling the AI, switches to a static flashcard interface (using pre-generated questions). The Student self-evaluates as correct/incorrect. 3. Continue step #11 Alternative flow 2: Student Pause the interview session 1. From #5 of the basic flow, if the Student selects the “Pause” option. 2. The system saves the current state (including conversation history, remaining turns) and exits the interface. (When the Student returns, the system will replay the question from Step #4) 3. The use case terminates Alternative flow 3: Student Skips a concept 1. From #4 or #5 of the basic flow, if the Student selects the “Skip” option. 2. The system immediately cancels the remaining returns and skips grading. 3. Continue step #10

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	Alternative flow 4: Student Appeals the grading result - From #9 of the basic flow, if after hearing the AI's feedback, the Student disagrees and selects "Appeals" - The system pauses the voice conversation flow, displays a form to enter the reason for the appeal. The concept's result is temporarily saved with and "Appealing" flag - Continue step #10
Pre-conditions	- The Student has logged in and completed the interview session configuration step - The list of concepts to review has been loaded into the queue by the system. - The system has a stable network connection to the AI service. - The Student has granted microphone access to the browser or device.
Post-conditions	- The comprehension score (mastery_score) of each concept tested in the session is fully updated in the database (concepts table). - If knowledge gaps are found, the foundational (prerequisite) concepts have been automatically inserted into the queue for the next study session via the Concept Graph Engine's algorithm. - The entire conversation history (text transcribed from speed and/or audio data) is successfully stored to serve as display data for the summary screen and future learning processes.

2.4 Use-case: Concept Graph Interaction

Use case Name	Concept Graph Interaction
Brief description	This use case allows Students to visually interact with the Concept Graph of their courses or study plans. By interacting through basic actions such as zoom, pan, hover tooltip, as well as searching and filtering concepts by their mastery score, Students can quickly grasp the overall picture of their knowledge gaps and the prerequisite relationships between concepts.
Actor	- Student
Basic Flow	Student selects the Concept Graph feature from the main Dashboard or through the main menu. System loads and displays the network diagram of concepts. The nodes (concepts) are colored based on the student's level of mastery (according to mastery_score). Student performs actions on the graph:

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	- Zoom: Zoom in or out to focus on a specific cluster of concepts or to view the entire graph. - Pan: Drag and move the viewport around the graph space. - Hover tooltip: Hover the mouse over a concept node. The system displays a tooltip with a brief summary, including the concept's name and its current mastery_score. 4. Student uses the toolbar to search and filter: - Enter a keyword in the search box to filter concepts by name. - Select a mastery level filter (e.g., weak, strong - corresponding to specific mastery_score ranges). 5. System automatically highlights the concepts that meet the search/filter criteria and dims the remaining nodes. 6. Student clicks on a concept node on the graph. 7. System displays a detail panel (sidebar/modal) for the selected concept, which includes: - Study history (History) of that concept across previous learning sessions. - A list of prerequisite concepts (Prerequisites) that must be mastered beforehand.
Alternative Flows	Alternative flow 1: No concepts found when filtering/searching 1. From step #5 of the basic flow, if the keyword or filter conditions do not match any concept in the graph. 2. The system displays the message "No matching concepts found", and the graph returns to its default display state (or dims all nodes). 3. Continue step #4. Alternative flow 2: Error loading graph data 1. From step #2 of the basic flow, if an error occurs while retrieving graph data from the database (network error or server error). 2. The system displays the error message "Unable to load the concept graph. Please try again later" and provides a "Try again" button so the student can retry the connection. 3. The use case terminates.
Pre-conditions	- The Student has successfully logged into the Recall AI system. - The Student has at least one study plan successfully extracted that contains a concept graph.
Post-conditions	- The system undergoes no data changes (this is a read-only task). The Student gains an overview of their knowledge mastery and the location of their weaknesses within the knowledge chain based on the visual graph.

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2.5 Use-case: Concept Traceback

Use case Name	Concept Traceback
Brief description	This use case describes the process where the system automatically traces the foundational knowledge gaps of a student when they answer incorrectly or receive an evaluation score below the threshold for a specific concept. The system utilizes a reverse Breadth-First Search (BFS) algorithm on the Concept Graph to identify weak prerequisites. This process runs entirely on software logic (purely Agentic, no AI API calls) to produce a list of concepts to review, allowing the student to verify and insert them into their upcoming study schedule.
Actor	- Scheduling & Remediation Engine (SRE) - Student
Basic Flow	1. Immediately after the system grades the Interview Session, if concept C has a mastery_score < threshold, the SRE automatically triggers the Concept Traceback feature. 2. The SRE initializes an empty queue Q. 3. The SRE queries the database (concept_edges) to retrieve all direct prerequisite concepts of C and enqueues them into Q. 4. The SRE iterates through the elements in queue Q. The traversal is constrained to a maximum depth of 2 levels (max_depth = 2) from C to prevent the subsequent study session from becoming excessively long. 5. For each prerequisite concept P dequeued from Q, the SRE checks the score history of P. The system determines if P has never been tested or if its mastery_score(P) < threshold. 6. If the condition in step #5 is met, the SRE inserts concept P at the front of the queue for the next study/review session, prioritizing learning P before returning to review C. Simultaneously, the prerequisite concepts of P (if the 2-level depth limit has not been exceeded) are enqueued to the end of Q. 7. After the traversal loop is complete, the system displays the newly found list of weak prerequisite concepts (prereqs) on the UI. 8. The student reviews the list of lacking foundational concepts and can then choose to confirm and add them to their study schedule, or skip one or more concepts. 9. The SRE saves the new schedule configuration based on the student's feedback and terminates the flow.
Alternative Flows	Alternative flow 1: Prerequisite Concept is Mastered 1. From step #5 of the basic flow, if the SRE detects that the prerequisite

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	concept P has been previously tested and has a mastery_score(P) >= threshold. - The SRE skips P (since this knowledge is already mastered and requires no review) and dequeues the next element in Q. The system will not traverse deeper into the prerequisites of this P. - Continue step #4. Alternative flow 2: No Prerequisites Found to Review - From step #4 of the basic flow, after completing the traversal of queue Q (queue is empty or max_depth reached), if the SRE finds no prerequisite concepts that are weak or untested. - The SRE treats concept C via the normal spaced repetition mechanism: it schedules a review for C after X days instead of an immediate review. - Continue step #9. Alternative flow 3: Student Skips Foundational Concepts - From step #8 of the basic flow, if the student decides to skip all or some of the foundational concepts proposed by the system. - The SRE removes the skipped concepts from the priority queue of the next session. The remaining concepts (if any) are still prioritized; for concept C, the system will schedule it for review as normal if all prerequisites are skipped. - Continue step #9.
Pre-conditions	- The concept graph for the study material/subject has been successfully created, including the concepts and the prerequisite relationships between them (concept_edges). - The student is in the middle of or has just completed an Interview Session. - The system detects a concept (referred to as concept C) that was answered incorrectly or has a mastery_score lower than the specified threshold (e.g., < 0.6).
Post-conditions	- The weak prerequisite concepts are successfully identified by the system. - The student's study schedule is updated: Foundational concepts needing reinforcement are inserted at the front of the priority queue for the next study session (to be studied before concept C). - Information about the weak prerequisites is stored so it can be displayed in detail in the End-of-Session Summary.